J.Sochor

Equivalent circuit method for 3-d magnetostatic calculations (ECM_MS)

Review of examples

A brief overview of examples of 3D magnetostatics calculations using the *ECM_MS* program is given. The *folder* contains files in *VoxCad* format (with the *.vxc* extension) with examples of source data.

The dimensions of the computational domain are provided, grid steps, number of contours (this is the number of faces in the computational grid, which is equal to the number of unknowns), number of branches (equal to the number of points for calculating magnetic flux density), number of cells (equal to the number of nodes in the computational grid), estimated computation time, information about the ferromagnetic material and the currents in the coils

Note. The examples provided are for research purposes; they are limited to the calculation of magnetic fluxes and are not intended for use in the design of industrial machines.

Contents

Part 1. Examples without comparisons 2
1.1 “core_big.vxc”. Magnetic core with large dimensions 2
1.2 “helices.vxc”. Arbitrary coil geometry and arbitrary magnetic core geometry 3
1.3 “Lens.vxc”. Solenoidal electromagnetic lens 3
1.4 “LSM.vxc”. Linear induction motor 4
1.5 “rotor_stator.vxc”. Cylindrical structure 5
1.6 “sphere.vxc”. Ferromagnetic sphere 5
1.6 “sphere_hollow.vxc”. Ferromagnetic hollow sphere 6
1.7 “step_motor.vxc”. Linear stepper motor 6
1.8 “Task13.vxc”. Similar to the benchmark task 13 7
Part 2. Examples with comparisons 8
2.1 “core_branched_asymm.vxc”. Closed branched magnetic core. Side coil 8
2.2. “core_branched_symm.vxc”. Closed branched magnetic core. Coil in the center 9
2.3. “core_simple.vxc” Closed magnetic core 10
2.4. “core_simple_4part.vxc” magnetic core composed of sections 12
2.5. “Task20.vxc” Similar to the benchmark task 20. 13
2.6. Constant magnetic permeability 15
Bibliography 16

Part 1. Examples without comparisons

1.1 “core_big.vxc”. Magnetic core with large dimensions

The magnetic core has a height of nearly 3 meters. Size of the computational domain: 2.5m × 2.35m ×3.4m. Grid steps: Δx=Δy=Δz=0.05m. Number of contours = 461673; number of branches = 470454; number of cells = 159800. Estimated computation time = 42s.

Materials:

Ferromagnet: Steel1020. The magnetization curve is shown in Fig. 1. The file *BH-ST20.txt* is located in the 'curves' folder. Coil: direct current 45000 AT.

Fig. 2a shows a screenshot, Fig. 2b displays the magnetic flux in a cross-section (the result is visualized in *ParaView*), and Fig. 2c presents a plot of the magnetic flux distribution along line *H*.

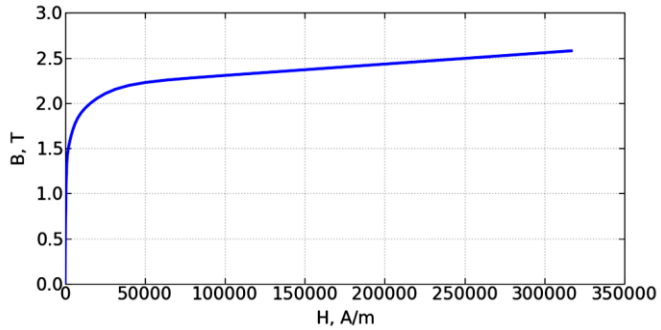
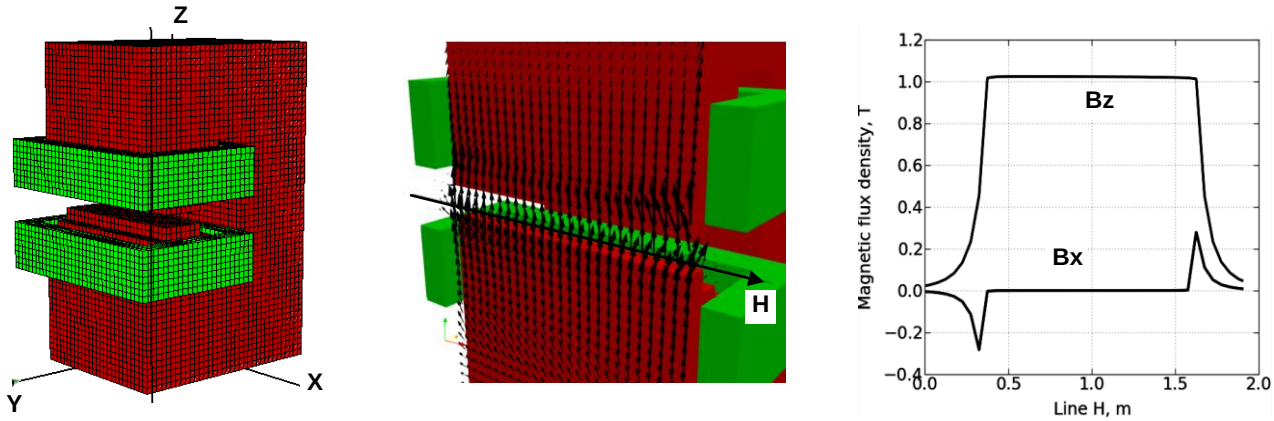


Fig 1. B-H curve graph (Steel1020)



a) b) c)

Fig. 2. a) screenshot of the structure; b) distribution of magnetic flux density in the cross section; c) graph of the distribution of the *Z*-component and *X*- component of the magnetic flux density along the *H*-line

1.2 “helices.vxc”. Arbitrary coil geometry and arbitrary magnetic core geometry

A helix coil wound around a helix magnetic core was used for the example.

Size of the computational domain: 0.18m × 0.148m ×0.156m. Grid steps: Δx=Δy=Δz= 0.004m. Number of contours = 189378; number of branches = 194195; number of cells = 66378. Estimated computation time = 10s

Materials: Ferromagnet μ_r = 1000. Coil - direct current 18 AT.

Fig. 1a shows are screenshot taken in *VoxCAD*, Fig. 1b result of calculations of the magnetic flux density field distribution in a ferromagnetic material and the current density in a winding



a) b)

Fig. 1. a) screenshot of the structure; b) result of calculations of the magnetic flux density field distribution in a ferromagnetic material and the current density in a winding

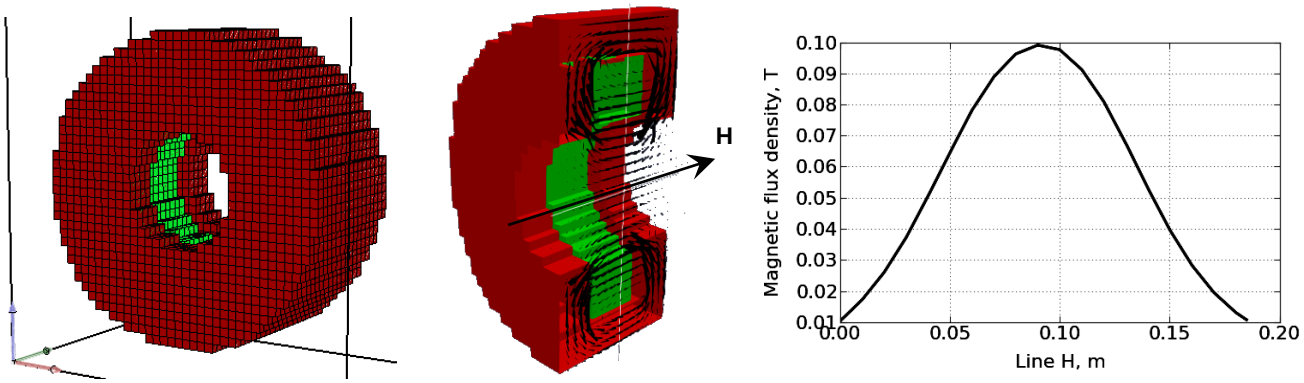
1.3 “Lens.vxc”. Solenoidal electromagnetic lens

Size of the computational domain: 0.44m × 0.24m ×0.44m. Grid steps: Δx=Δy=Δz=0.01m. Outer diameter 0.35 m, inner diameter 0.13 m, length of the yoke along the lens axis 0.16 m, coils thickness 0.06 m.

Number of contours = 131408; number of branches = 135344; number of cells = 46464.

Estimated computation time = 10s. Materials: Ferromagnet Steel1020. The magnetization curve is shown in Fig. 1 in section 1.1. Coil: direct current 9600 AT.

A specific aspect of the calculation is that it was necessary to set a gap (0.01m) between the plane of the coil and the yoke. Fig. 1a shows a screenshot, Fig. 1b displays the magnetic flux in a cross-section (the result is visualized in *ParaView*), and Fig. 1c presents a plot of the magnetic flux distribution along line *H*.



a) b) c)

Fig. 1. a) screenshot of the structure; b) distribution of magnetic flux density in the cross section; c) graph of the distribution of the *Y*-component of the magnetic flux density along the *H*-line

1.4 “LSM.vxc”. Linear induction motor

Linear machines have a simple design that can be easily modeled using cuboids. This allows for the evaluation of mathematical and software solutions by performing calculations for various designs.

A three-phase winding is placed in a stationary magnetic core and creates a moving magnetic field, first in the positive **x**-axis direction, then in the opposite direction. A direct current is applied to the excitation winding, and it moves synchronously with the field of the three-phase winding. The velocity of the field **Vs** of the three-phase winding is **2·τ·fs**, where **τ** is the pole pitch, **fs** is the frequency of the three-phase winding current.

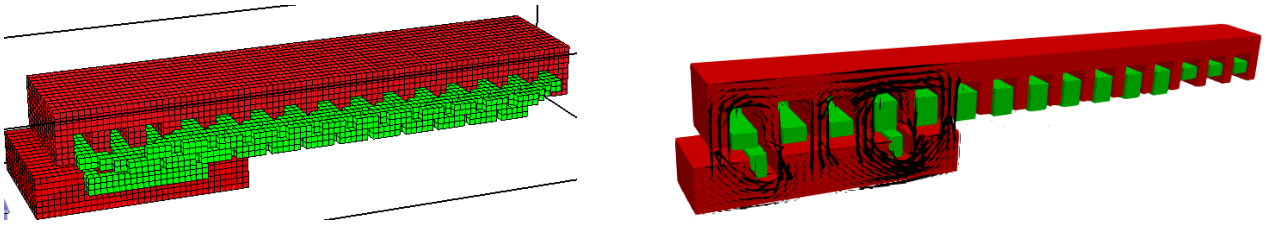
Size of the computational domain: 0.2875m × 0.10875m ×0.0575m. Grid steps: Δx=0.0025m, Δy=0.00375, Δz=0.0025m. Number of contours = 216988; number of branches = 223468; number of cells = 76705. Δt=0.001s

The estimated computation time is 1.3 s per step Δt.

Materials: Ferromagnet μ_r = 100.

Coil: in a three-phase winding, a current with an amplitude of 50AT and a frequency fs=28Hz. Direct current in the field winding 50 AT

Fig. 1a shows are screenshot taken in *VoxCAD*, Fig. 1b result of calculations of the magnetic flux density field distribution in a ferromagnetic material



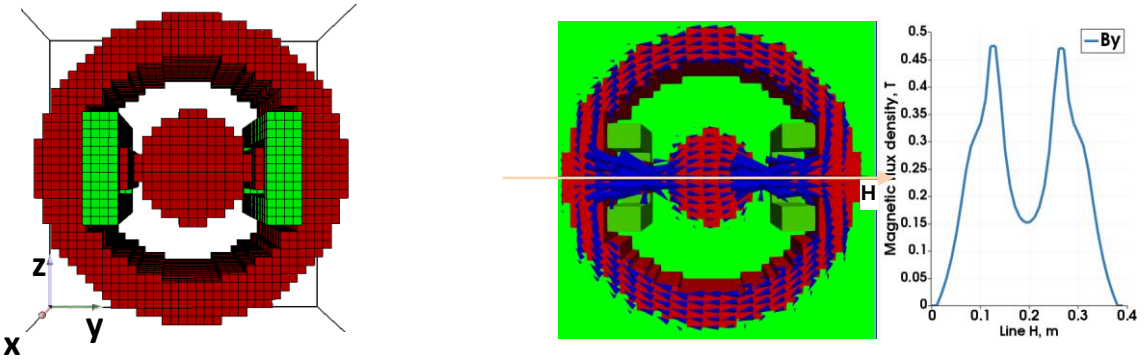
a) b)

Fig. 1. a) screenshot of the structure; b) distribution of magnetic flux density in the cross section

1.5 “rotor_stator.vxc”. Cylindrical structure

Size of the computational domain: 0.44m × 0.44m ×0.44m. Grid steps: Δx=Δy=Δz=0.01m. Number of contours = 244068; number of branches = 249744; number of cells = 85184. Estimated computation time: 12s.

Materials: Ferromagnet μ_r = 200. Coil: direct current 1000 AT.



a) b) c)

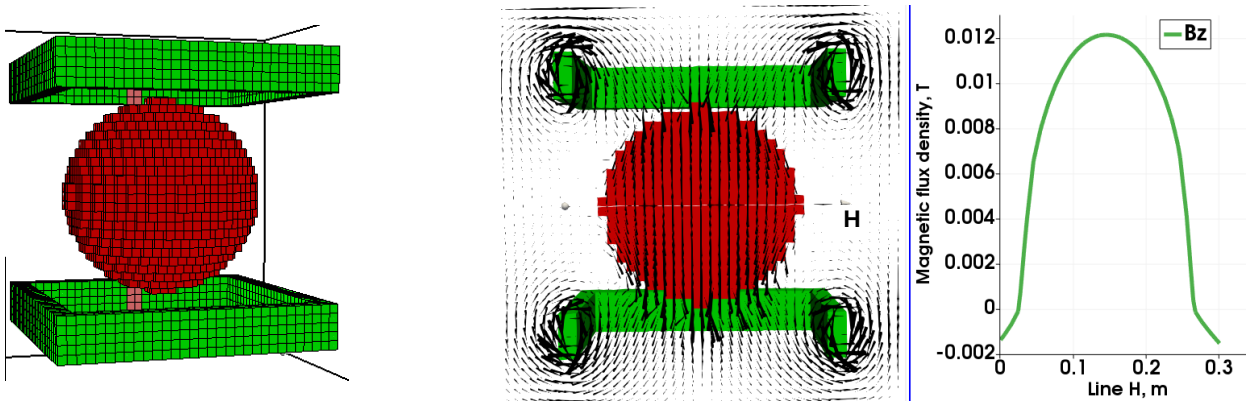
Fig. 1. a) screenshot of the structure; b) distribution of magnetic flux density in the cross section; c) graph of the distribution of the *Y*-component of the magnetic flux density along the *H*-line.

A specific feature: it was necessary to provide a gap between the winding and the ferromagnetic material.

1.6 “sphere.vxc”. Ferromagnetic sphere

Size of the computational domain: 0.44m × 0.44m ×0.44m. Grid steps: Δx=Δy=Δz=0.01m. Number of contours = 244068; number of branches = 249744; number of cells = 85184. Estimated computation time: 16s.

Materials: Ferromagnet μ_r = 200. Coil: direct current 5000 AT.



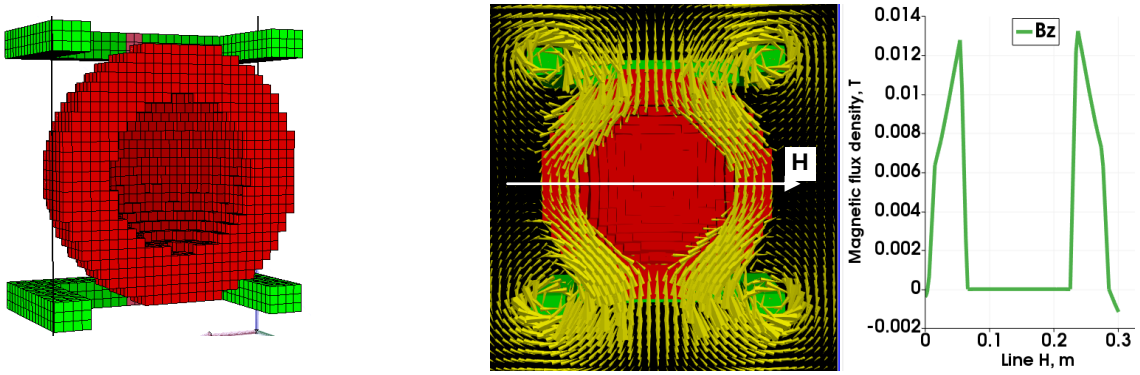
a) b) c)

Fig. 1. a) screenshot of the structure; b) distribution of magnetic flux density in the cross section; c) graph of the distribution of the *Z*-component of the magnetic flux density along the H-line.

1.6 “sphere_hollow.vxc”. Ferromagnetic hollow sphere

Size of the computational domain: 0.44m × 0.44m ×0.44m. Grid steps: Δx=Δy=Δz=0.01m. Number of contours = 244068; number of branches = 249744; number of cells = 85184. Estimated computation time: 16s.

Materials: Ferromagnet μ_r = 200. Coil: direct current 5000 AT.



a) b) c)

Fig. 1. a) screenshot of the structure (cross section); b) distribution of magnetic flux density in the cross section; c) graph of the distribution of the *Z*-component of the magnetic flux density along the *H*-line.

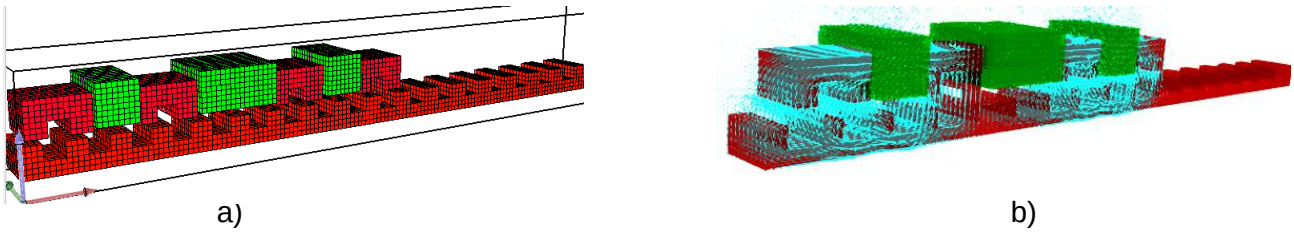
1.7 “step_motor.vxc”. Linear stepper motor

In this version of a linear stepper motor, current pulses are applied to windings located on the moving part. This generates sequences of magnetic flux pulses that interact with the stationary part of the motor, which features a toothed structure. As a result, the motor undergoes discrete movement by precisely defined distances - steps.

Size of the computational domain: 0.775m × 0.1m ×0.11m. Grid steps: Δx=Δy=Δz=0.005m. Number of contours = 184712; number of branches = 191260; number of cells = 66000.
The estimated computation time is 1.25 s per step Δt.

Materials: Ferromagnet μ_r = 200.

Coil: Current pulse amplitude - 20 AT. Direct current in the magnetizing coil 5 AT



a) b)

Fig. 1. a) screenshot of the structure; b) distribution of magnetic flux density

1.8 “Task13.vxc”. Similar to the benchmark task 13

This example is similar to the benchmark task 13 for the TEAM workshop [1] [2].
Size of the computational domain: 0.28m × 0.224m ×0.152m. Grid steps: Δx=Δy=Δz=0.004m.
Number of contours = 429628; number of branches = 438172; number of cells = 148960.
Estimated computation time: 61s.
Ferromagnet: the magnetization curve is shown in Fig. 1 [2]. The file *BH-team13.txt* is located in the 'curves' folder.
Coil: direct current 1000 AT.

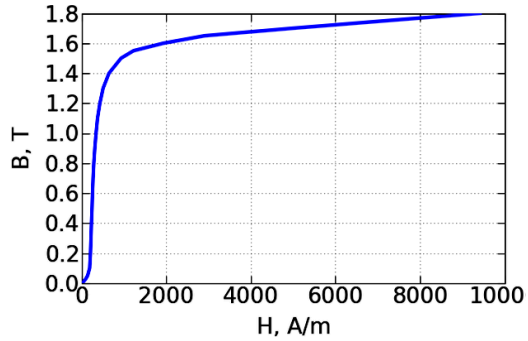
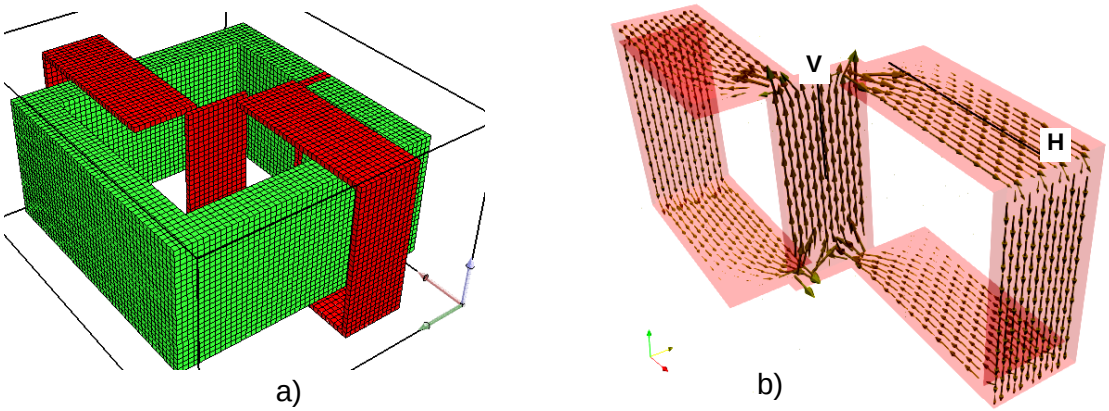
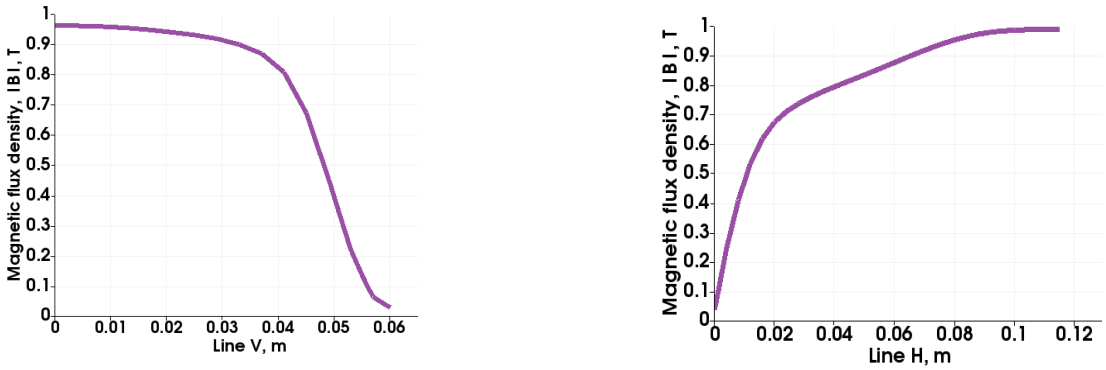


Fig 1. B-H curve of steel [2]



a) b)

Fig. 2. a) screenshot of the structure; b) distribution of magnetic flux density



a) b)

Fig. 3. a) graph of the distribution of the of the magnetic flux density along the *V* line.
b) graph of the distribution of the of the magnetic flux density along the *H* line

Part 2. Examples with comparisons

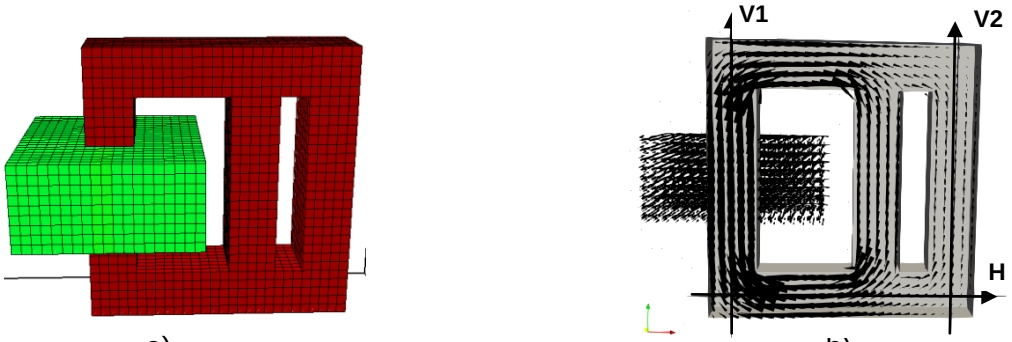
2.1 “core_branched_asymm.vxc”. Closed branched magnetic core. Side coil

Size of the computational domain: 0.106m × 0.076m ×0.078m. Grid steps: Δx=Δy=Δz=0.002m.

Number of contours = 224642; number of branches = 230075; number of cells = 78546.

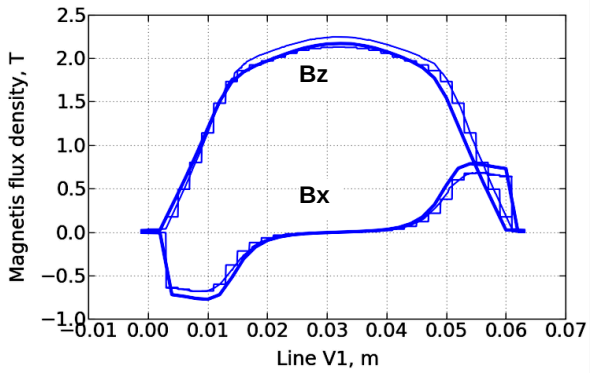
Estimated computation time: 12s.

Materials: Ferromagnet μ_r = 100. Coil: direct current 1800 AT

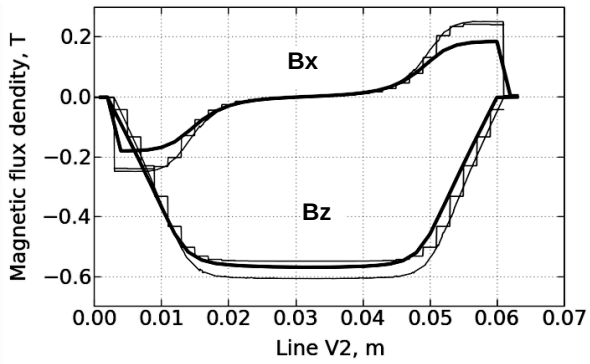


a) b)

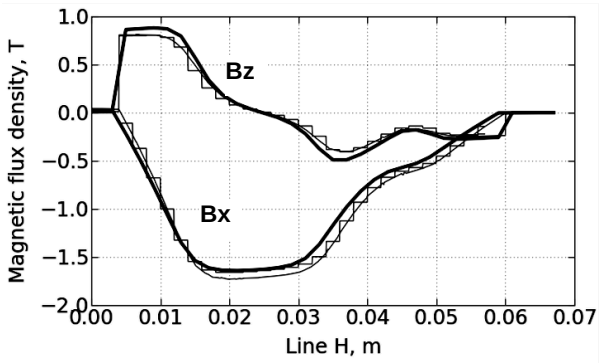
Fig. 1. a) screenshot of the structure; b) distribution of magnetic flux density in the cross section;



a)



b)



c)

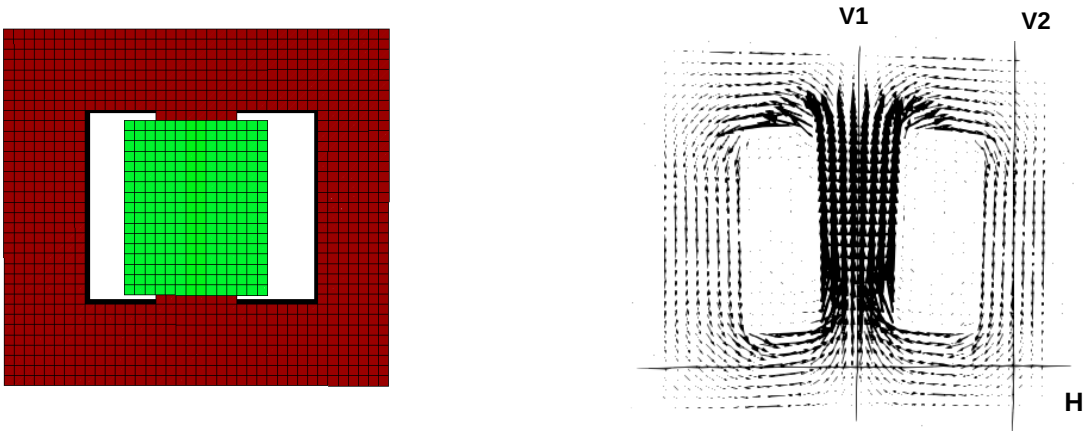
Fig.2. a), b), c) – plots of the magnetic flux density distribution along lines *V1*, *V2*, and *H*, respectively.

Thickened line correspond to calculations in the *ECM_MS*, thin line correspond to calculations in the *FEM,* stepped lnes correspond to calculations in the *PyPEEC* software

2.2. “core_branched_symm.vxc”. Closed branched magnetic core. Coil in the center

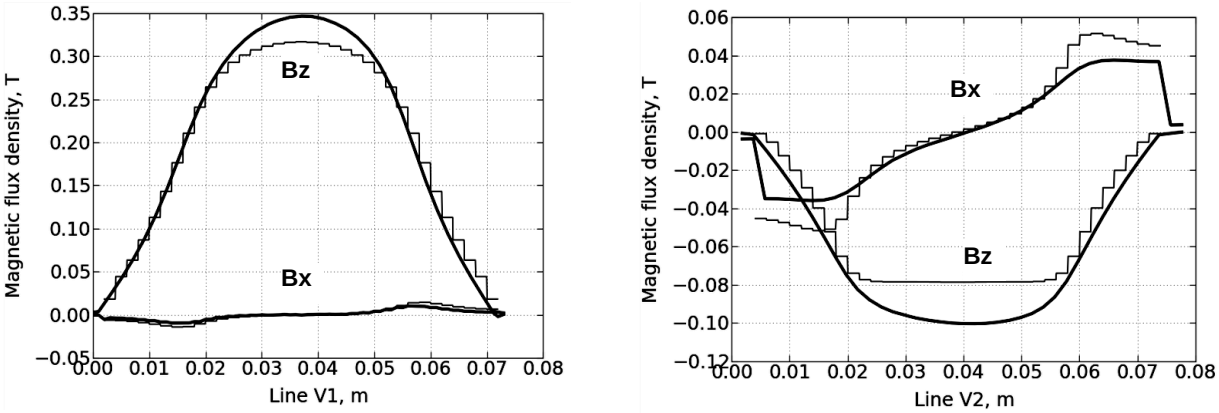
Size of the computational domain: 0.106m × 0.076m ×0.078m. Grid steps: Δx=Δy=Δz=0.002m. Number of contours = 224642; number of branches = 230075; number of cells = 78546.

Estimated computation time: 12s. Materials: Ferromagnet μ_r = 100. Coil: direct current 1800 AT

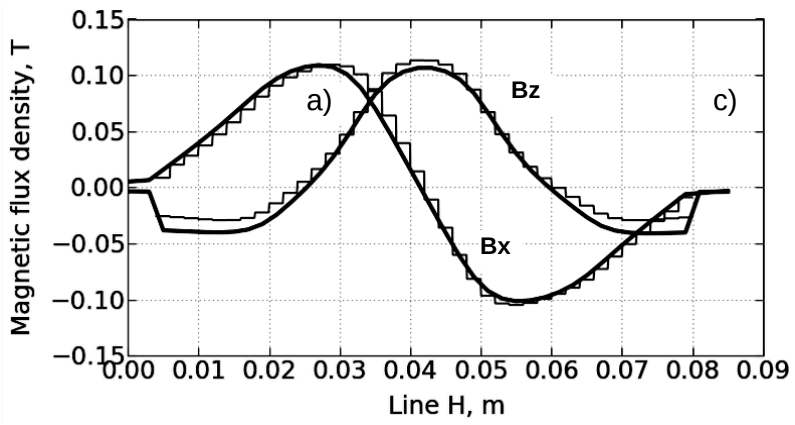


a) b)

Fig. 1. a) screenshot of the structure; b) distribution of magnetic flux density in the cross section;



a) b)



c)

Fig.2. a) a), b), c) – plots of the magnetic flux density distribution along lines *V1*, *V2*, and *H*, respectively. Thickened line correspond to calculations in the *ECM_MS*, stepped lnes correspond to calculations in the *PyPEEC* software

2.3. “core_simple.vxc” Closed magnetic core

Size of the computational domain: 0.106m × 0.076m ×0.078m. Grid steps: Δx=Δy=Δz=0.002m. Number of contours = 224642; number of branches = 230075; number of cells = 78546.

Materials: Ferromagnet: the magnetization curve is shown in Fig. 1 [3]. The file BH-team20.txt is located in the 'curves' folder. Coil: direct current 1800 AT

Estimated computation time: 39s.

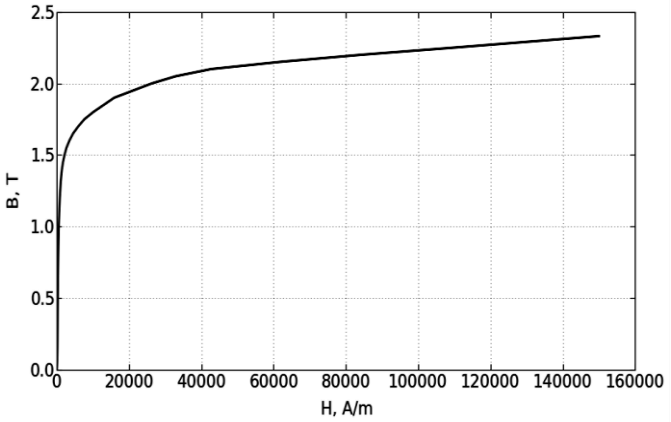
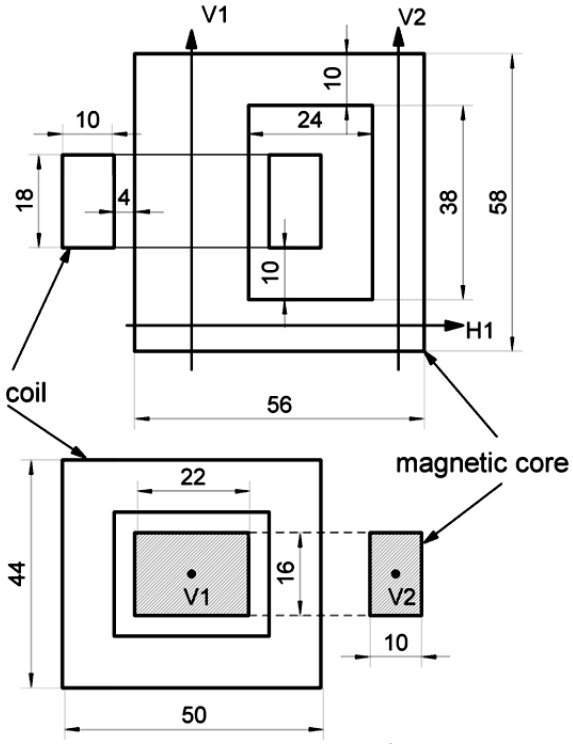
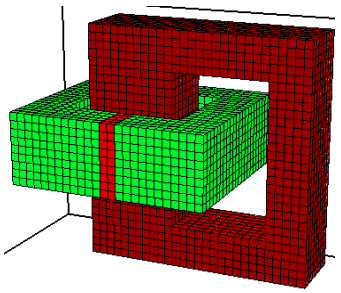


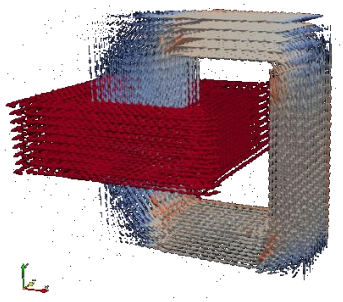
Fig 1. B-H curve graph [3]



a)



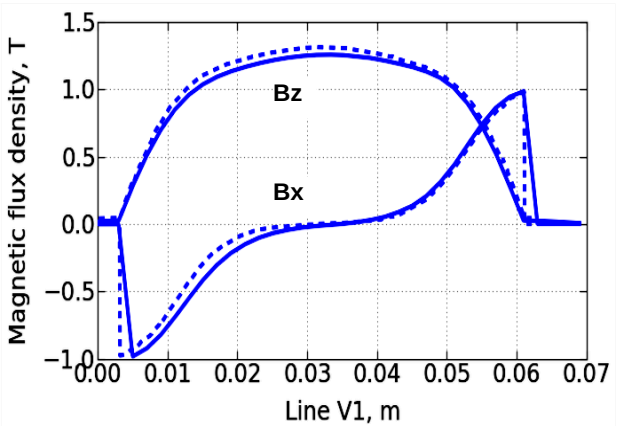
b)



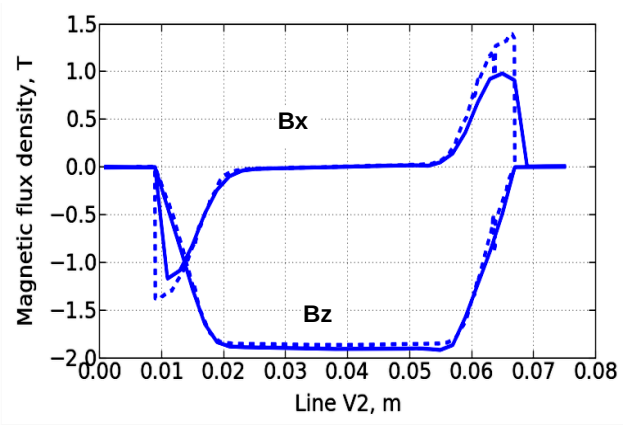
c)

Fig.2. a) the domain sizes, control lines: *V1*, *V2*, *H1*;
b) 3d image of the structure.
c) coil current density field and magnetic flux density vector field calculated in *ECM_MS*.

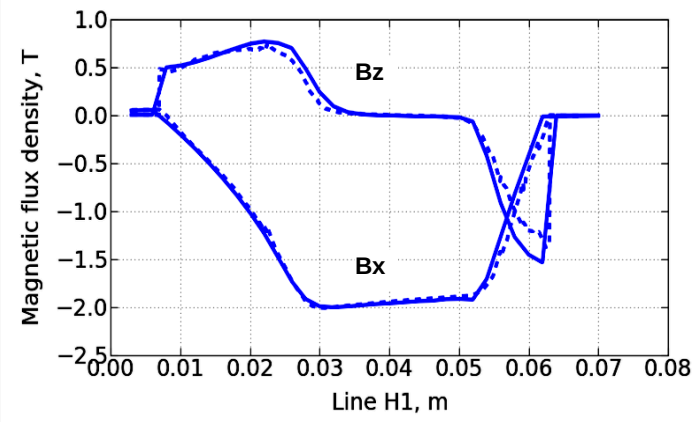
Below in Fig.3 are graphs of the *X*- *Z*- components of the magnetic flux density along the control lines. The positive directions along the control lines are shown in Fig.2a



a) Magnetic flux density along line *V1*



b) magnetic flux density along line *V2*

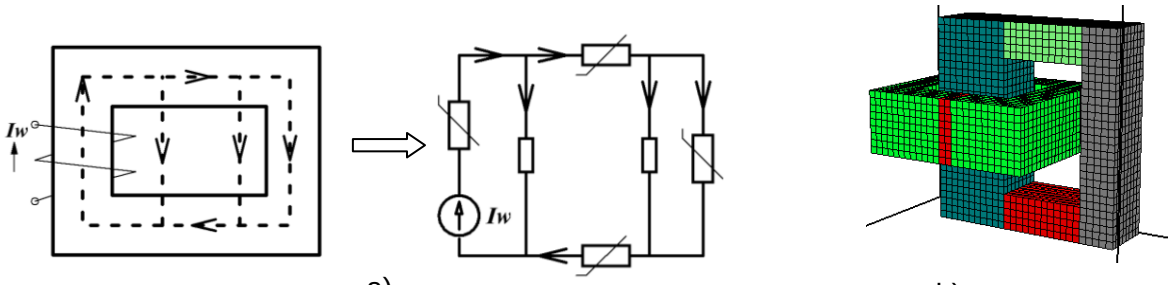


c) magnetic flux density along line *H1*

Fig.3. Graphs of the magnetic flux density. Solid lines of the graphs correspond to calculations in the *ECM_MS*, dotted lines correspond to calculations in the *FEM* software

2.4. “core_simple_4part.vxc” magnetic core composed of sections

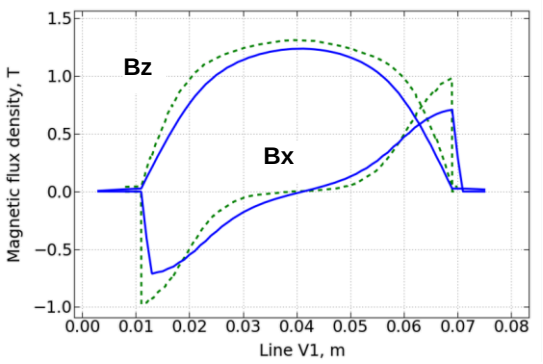
Sometimes, when performing engineering calculations, the computational domain is subdivided into sections that are represented as equivalent magnetic reluctances (Fig. 1a). In the EMC_MS software, this approach corresponds to a calculation mode in which the ferromagnetic material is also divided into sections, with its nonlinear properties averaged over the volume of each section. In Fig. 1b, these sections are highlighted in different colors.



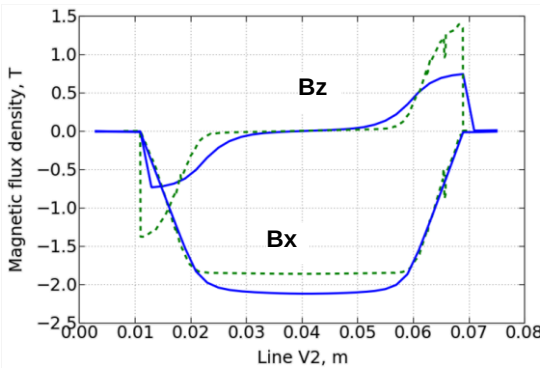
a) b)

Fig. 1. In approximate calculations, the ferromagnet is divided into sections

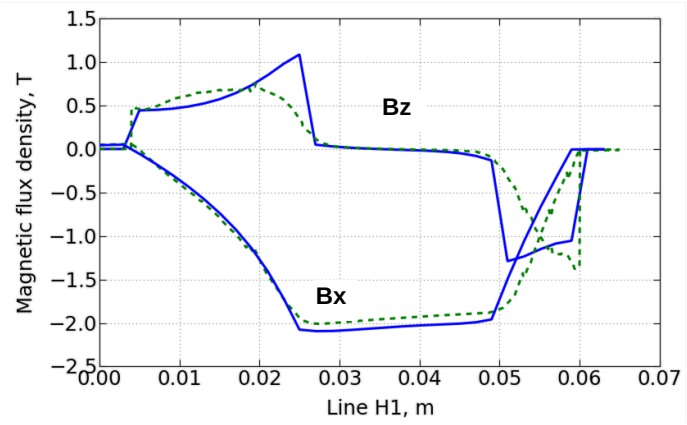
Below in Fig.2 are graphs of the *X*- *Z*- components of the magnetic flux density along the control lines. The positive directions along the control lines are shown in Fig.2a from section 2.3



a) Magnetic flux density along line *V1*



b) magnetic flux density along line *V2*



c) magnetic flux density along line *H1*

Fig.9. Graphs of the magnetic flux density. Solid lines of the graphs correspond to calculations in the *ECM_MS*, dotted lines correspond to calculations in the FEM software

The estimated time is cut in half — to 20 sec (in mode=3)

2.5. “Task20.vxc” Similar to the benchmark task 20.

This example is similar to the benchmark task 20 for the TEAM workshop [3]. In addition to the variable cross-section, the magnetic core has contacts along the edges. Figure 1.1 shows the dimensions of the computational domain.

To evaluate the saturation effect, the coil ampere-turn values of 250, 500, 1000, 3000, 4500 and 5000 (direct current) were selected. The calculated quantities are the following: the average values of the ***z***-component of the magnetic flux density along the α-β line in the central strip and the γ-δ line in the yoke, and also along the **z**-axis in the central part of the yoke.

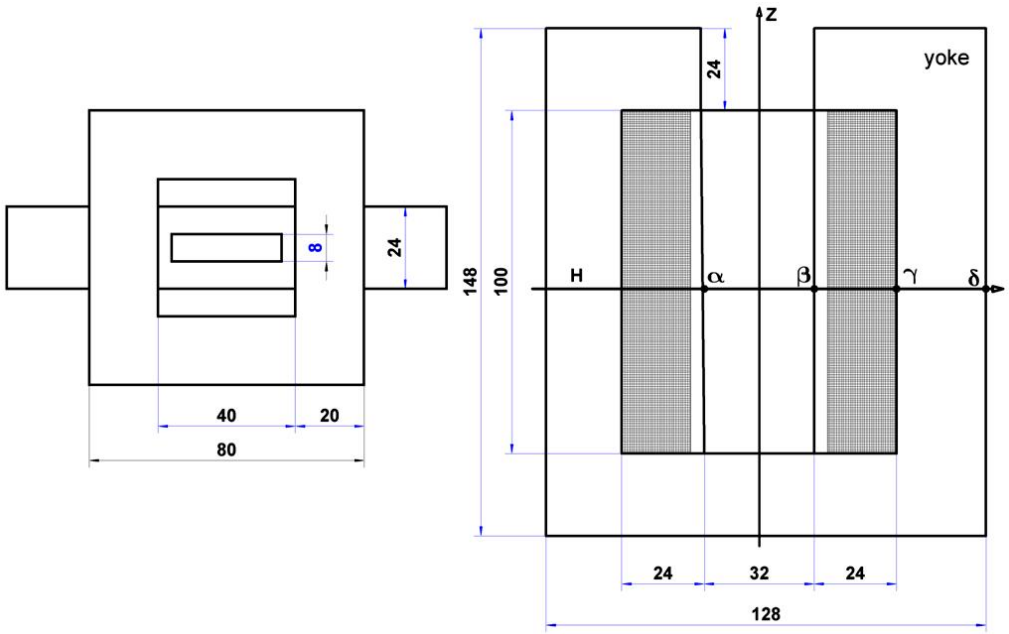


Fig 1. Shows the domain sizes for the test 1 example.
The first control line is *H*; segments α–β and γ–δ lie on it. The second control line coincides with the *Z*-axis..

Fig. 2 shows the B-H curve for steel [3], [4].

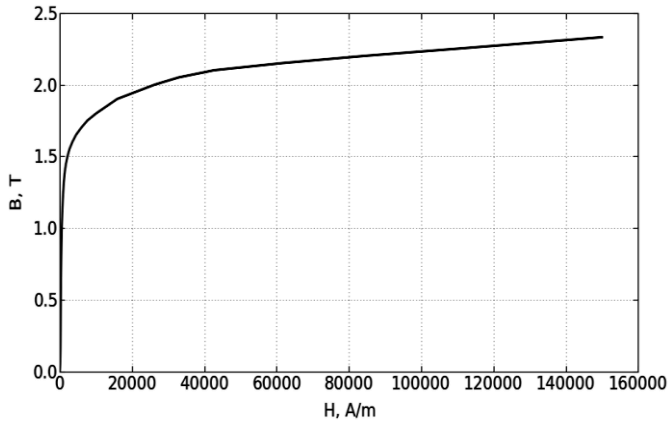
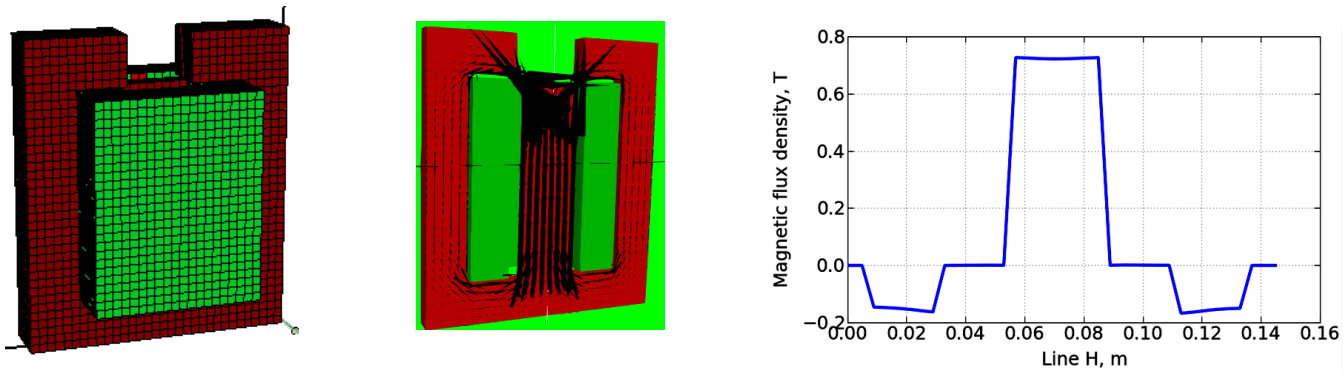


Fig 2. B-H curve graph

Below in fig. 3a shows a screenshot of the structure, prepared in a graphical editor. Size of the computational domain: 0.176m × 0.096m ×0.192m. Grid steps: Δx=Δz=0.004m, Δy=0.002m. Number of contours =291212; number of branches =297600; number of cells=101376.



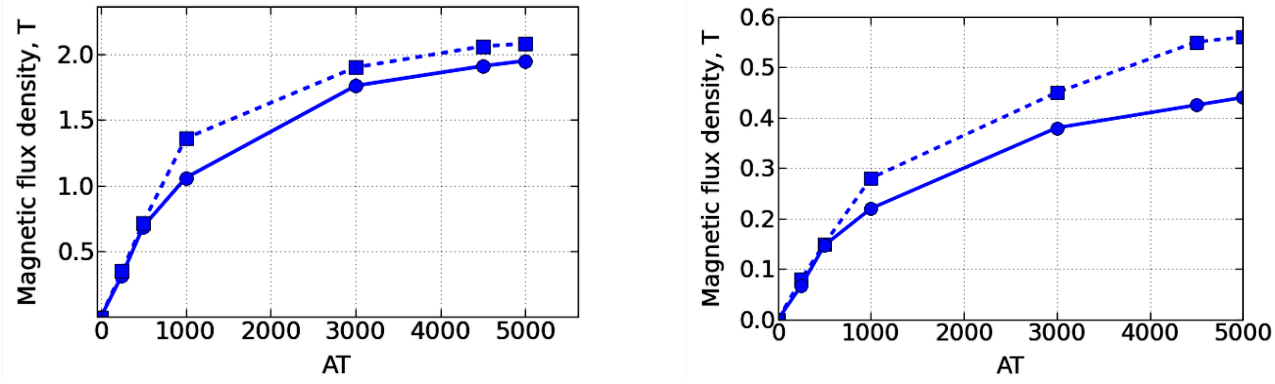
a) b) c)

Fig. 3. a) screenshot of the structure; b) distribution of magnetic flux density in the cross section; c) graph of the distribution of the Z-component of the magnetic flux density along the H-line at a current in the coil of 500 AT

The calculation results obtained using the *ECM_MS* system were compared with those obtained using a program based on FEM algorithms. Table 2 and Fig.1.4 presents the values of the **z**-components of the average flux densities in the central band (α−β) and yoke (γ−δ) based on the results of calculations in *ECM_MS* and and *FEM* software. The table also shows the calculation time values.

Table 2 Z-components of average flux densities in center strip (α-β), yoke (γ-δ) and calculation time

pos.	soft	unit	250AT	500AT	1000AT	3000AT	4500AT	5000AT
α-β	ECM	T	0.35	0.71	1.12	1.76	1.9	1.94
	FEM		0.35	0.72	1.39	1.9	2.05	2.08
γ-δ	ECM	T	0.074	0.147	0.24	0.38	0.43	0.45
	FEM		0.08	0.15	0.28	0.45	0.55	0.56
time	ECM	sec	69	74	82	82	82	84
		tol iter	2.5e-4 330	2e-4 370	1.2e-4 420	1e-5 464	2e-6 470	2e-6 489
	FEM	sec	1560	1740	1020	1200	1560	2200



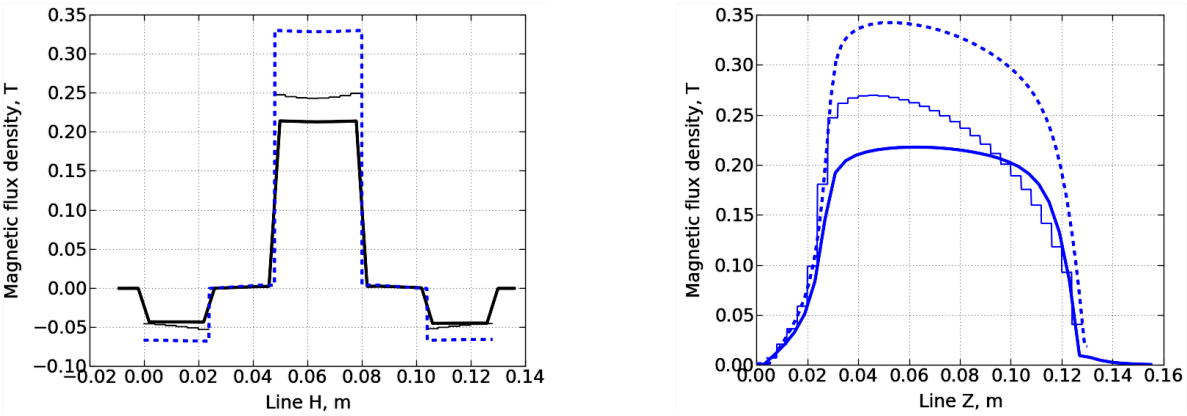
a) segment α-β b) segment γ- δ

Fig. 1.4 The *Z*-component of average flux density in the central strip (α−β) and yoke (γ−δ). Solid lines in the graphs correspond to calculations in *ECM_MS*, dotted lines - in *FEM*.

2.6. Constant magnetic permeability

A software package *PyPEEC* was additionally used for comparative calculations. Since *PyPEEC* does not support nonlinear elements, the calculations were performed assuming constant relative magnetic permeability.

Fig. 1 presents the calculation results for the previous example, but with a constant value of μ_r=100. Coil: direct current 500 AT The positions of the reference lines *H* and *Z* are indicated in Fig.1 in the preceding section.



a) flux density along the *H*-line

b) flux density along the *Z*-line

Fig. 1.6 The *Z*-component flux density.
Solid lines in the graphs correspond to calculations in *ECM_MS*, dotted lines - in *FEM*, stepped lnes correspond to calculations in the *PyPEEC* software.

Bibliography

[1]. Team Problem 13. 3-D Non-Linear Magnetostatic Model. URL: https://compumag.org/jsite/images/stories/TEAM/problem13.pdf

[2]. O. Biro, “SOLUTION OF TEAM BENCHMARK PROBLEM #13 (3-D NONLINEAR MAGNETOSTATIC MODEL)”, *ACES Journal*, vol. 8, no. 2, pp. 216–225, Jul. 2022.

[3]. Team Problem 20. International Compumag Society. URL: https://compumag.org/jsite/images/stories/TEAM/problem20.pdf

[4]. N. Takahashi, T. Nakata and H. Morishige" Summary of Results for Problem 20 (3-D Static Force Problem)". *COMPEL*, 1995. Vol. 14 No. 2-3 pp. 57–75, doi: https://doi.org/10.1108/eb010138

---

november, 2026

Website